

Playing Around with Graphs in Maxima

In the final and 24th article in this series, the reader gets to play around with graphs using previously gained knowledge about the graphs package of Maxima.

In the previous article in this series, we got familiar with simple graphs, and how the graphs package of Maxima allows us to create and visualise them. Building on that knowledge, in this article, we are going to play around with graphs and their properties, using the functions provided by Maxima's graphs package.

Graph modifications

We have already created various graphs with the `create_graph()` and `make_graph()` functions of the graphs package of Maxima. What if we wanted to modify some existing graphs, say by adding or removing some edges or vertices? For such operations, Maxima provides the following functions:

- `add_edge(<edge>, <g>)` - Adds <edge> into the graph <g>
- `add_edges(<edge_list>, <g>)` - Adds edges specified by <edge_list> into the graph <g>
- `add_vertex(<vertex>, <g>)` - Adds <vertex> into the graph <g>
- `add_vertices(<vertex_list>, <g>)` - Adds vertices specified by <vertex_list> into the graph <g>
- `connect_vertices(<u_list>, <v_list>, <g>)` - Connects all vertices from <u_list> to all vertices in <v_list> in the graph <g>
- `contract_edge(<edge>, <g>)` - Merges the vertices of the <edge> and the edges incident on those vertices, in the graph <g>
- `remove_edge(<edge>, <g>)` - Removes the <edge> from the graph <g>
- `remove_vertex(<vertex>, <g>)` - Removes the <vertex> and the associated edges from the graph <g>

Some of the above functions are demonstrated below:

```
$ maxima -q
(%i1) load(graphs)$ /* Loading the graphs package */
...
0 errors, 0 warnings
(%i2) g: create_graph(4, [[0, 1], [0, 2]]);
(%o2) GRAPH(4 vertices, 2 edges)
(%i3) print_graph(g)$
```

Graph on 4 vertices with 2 edges.

```
Adjacencies:
3 :
2 : 0
1 : 0
0 : 2 1
(%i4) add_edge([1, 2], g)$
(%i5) print_graph(g)$
```

Graph on 4 vertices with 3 edges.

```
Adjacencies:
3 :
2 : 1 0
1 : 2 0
0 : 2 1
(%i6) contract_edge([0, 1], g)$
(%i7) print_graph(g)$
```